

Q3
Selecting Exception Using
Exception Propagation
10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[] a = new int[5];
6         a[0] = 10;
7         a[1] = 20;
8         a[2] = 30;
9         a[3] = 40;
10        a[4] = 50;
11        int index = sc.nextInt();
12        try {
13            System.out.println(a[index]);
14        } catch (ArrayIndexOutOfBoundsException e) {
15            System.out.println("Exception: Invalid array index.");
16        }
17    }
18 }
```

Submitted

Input

6

Output

Exception: Invalid array index.

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

10.00 / 10 marks

Q4

Select Validation
10 marks

Q5

Selecting Exception Using
Exception Propagation
10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int marks = sc.nextInt();
6         try {
7             if (marks < 0 || marks > 100) {
8                 throw new IllegalArgumentException();
9             }
10            System.out.println("Valid marks: " + marks);
11        } catch (IllegalArgumentException e) {
12            System.out.println("Exception: Invalid marks. Marks should be between 0 and 100.");
13        }
14    }
15 }
```

Submitted

Input

120

Output

Exception: Invalid marks. Marks should be between 0 and 100.

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

10.00 / 10 marks

- Q4
Final Validation
10 marks
- Q5
Handling Exception using
Exception Propagation
10 marks
- 5/5 answered

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner scanner = Scanner(System.in);
5         String password=sc.nextLine();
6         try{
7             if(password.length()<8){
8                 throw new WeakPasswordException(
9                     "Password must contain at least 8 characters."
10                );
11             }
12             System.out.println("Password is strong.");
13             throw (WeakPasswordException) null;
14             System.out.println("Exception: " + e.getMessage());
15         }
16     }
17 }
18 class WeakPasswordException extends Exception {
19     WeakPasswordException(String message) {
20         super(message);
21     }
22 }
```

Submitted

Input

java123

Output

Exception: Password must contain at least 8 characters.

Visible Test Cases

- Test Case 1 
- Test Case 2 
- Test Case 3 

10.00 / 10 marks

- Q4
Final Validation
10 marks
- Q5
Handling Exception using
Exception Propagation
10 marks
- 5/5 answered

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner scanner = Scanner(System.in);
5         String email=sc.nextLine();
6         try{
7             if(!email.matches(".*")){
8                 throw new InvalidEmailException("Invalid email address");
9             }
10            System.out.println("Valid email address.");
11            throw (InvalidEmailException) null;
12            System.out.println("Exception: " + e.getMessage());
13        }
14    }
15 }
16 class InvalidEmailException extends Exception {
17     InvalidEmailException(String message) {
18         super(message);
19     }
20 }
```

Submitted

Input

student@gmail.com

Output

Exception: Invalid email address.

Visible Test Cases

- Test Case 1 
- Test Case 2 
- Test Case 3 

10.00 / 10 marks

Your Code

```
1 import java.util.*;
2
3
4
5
6 public class Main {
7
8     static int balance;
9
10    static void withdraw(int amount) throws InsufficientBalanceException {
11        validateBalance(amount);
12    }
13
14    static void validateBalance(int amount) throws InsufficientBalanceException {
15        if (amount > balance) {
16            throw new InsufficientBalanceException();
17            "Withdrawal amount exceeds available balance."
18        }
19    }
20
21    static void processTransaction(int amount) throws InsufficientBalanceException {
22        withdraw(amount);
23    }
24
25    public static void main(String[] args) {
26
27        Scanner sc = new Scanner(System.in);
```

Input

Balance:
5000
Withdraw:
7000

Output

Exception propagated to main(),
InsufficientBalanceException:
Withdrawal amount exceeds available
balance.

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

10.00 / 10 marks